

Chapter 16: Oscillatory Motion and Waves

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PHY201
Lecture 16



Part I

Simple Harmonic Motion

Outline of Part I: Simple Harmonic Motion

Introduction to Oscillatory Motion

Hooke's Law

Period and Frequency

Simple Harmonic Motion

The Simple Pendulum

Energy in SHM

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What is Oscillatory Motion?

Oscillatory Motion

Oscillatory motion is motion that repeats itself regularly about an **equilibrium position**.

Everyday Examples

- ▶ A child on a swing
- ▶ A mass bouncing on a spring
- ▶ The pendulum of a grandfather clock
- ▶ Atoms vibrating in a solid
- ▶ Sound waves in air
- ▶ Tides in the ocean

All of these share a common feature:

A **restoring force** pulls the object back toward equilibrium whenever it is displaced.

The resulting back-and-forth motion is called an **oscillation** (or **vibration**).

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Hooke's Law: Restoring Force

Hooke's Law — SI Unit of k : newton per meter ($\frac{\text{N}}{\text{m}}$)

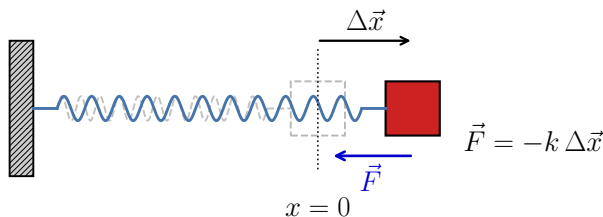
When a spring (or other elastic object) is displaced from equilibrium by an amount $\Delta\vec{x}$, it exerts a **restoring force**:

$$\vec{F} = -k \Delta\vec{x}$$

where k is the **force constant** (or spring constant) and the minus sign indicates the force opposes the displacement.

Force Constant k

- ▶ Large k : stiff spring, large restoring force.
- ▶ Small k : weak spring, small restoring force.
- ▶ k depends on material and geometry.



Restoring force opposes displacement.

Elastic Potential Energy

Elastic Potential Energy PE_{el} — SI Unit: joule (J)

When a spring is compressed or stretched by x , work is done and stored as **elastic potential energy**:

$$PE_{el} = \frac{1}{2}kx^2$$

- ▶ Work done to compress/stretch a spring equals $\frac{1}{2}kx^2$.
- ▶ This energy is **stored** in the spring and released when the force is removed.
- ▶ $PE_{el} \geq 0$ regardless of the sign of x .
- ▶ This is the same concept as gravitational PE: energy stored in a configuration.

$$PE_{el} = \frac{1}{2}kx^2$$

At maximum displacement
($x = X$):

$$PE_{el} = \frac{1}{2}kX^2 = \text{total energy}$$

Example: Car Suspension Spring — (cp2e_ch16_ex1)

What is the force constant for a car suspension that settles 1.20 cm when an 80.0 kg person gets in?

Recognize:

$$\Delta x = -1.20 \times 10^{-2} \text{ m}$$

$$\begin{aligned} F &= mg = (80.0 \text{ kg})(9.80 \frac{\text{m}}{\text{s}^2}) \\ &= 784 \text{ N} \end{aligned}$$

($\vec{F}$ upward, $\Delta \vec{x}$ downward $\Rightarrow$ opposite signs)

$$F = -k \Delta x$$

$$k = -\frac{F}{\Delta x} = -\frac{784 \text{ N}}{-1.20 \times 10^{-2} \text{ m}}$$

$$k = 6.53 \times 10^4 \frac{\text{N}}{\text{m}}$$

Without shock absorbers, the car would oscillate at this k .

Example: Toy Gun Spring — (cp2e_ch16_ex2)

A toy gun spring has $k = 50.0 \frac{\text{N}}{\text{m}}$ and is compressed $x = 0.150 \text{ m}$.

(a) How much energy is stored? (b) At what speed does a 2.00 g projectile leave?

$$\begin{aligned} \text{(a)} \quad PE_{el} &= \frac{1}{2} k x^2 \\ &= \frac{1}{2} (50.0 \frac{\text{N}}{\text{m}}) (0.150 \text{ m})^2 \end{aligned}$$

$$PE_{el} = 0.563 \text{ J}$$

$$\text{(b)} \quad \frac{1}{2} m v^2 = PE_{el}$$

$$v = \sqrt{\frac{2 PE_{el}}{m}} = \sqrt{\frac{2(0.563 \text{ J})}{2.00 \times 10^{-3} \text{ kg}}}$$

$$v = 23.7 \frac{\text{m}}{\text{s}}$$

$$23.7 \frac{\text{m}}{\text{s}} \approx 85 \frac{\text{km}}{\text{h}}$$

The spring stores energy; the projectile converts it all to kinetic energy.

We used **energy conservation**:

$$KE_f = PE_{el,i}$$

(frictionless, massless spring)

Check Your Understanding

A spring with $k = 200 \frac{\text{N}}{\text{m}}$ is compressed by 0.10 m.

(a) How much elastic potential energy is stored in the spring?

(b) If this energy is used to launch a 50 g ball, what is its launch speed?

Come to lecture to see the answer.

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Period and Frequency of Oscillation

Period T — SI Unit: second (s)

The time required to complete **one full oscillation** (one cycle).

$$T = \frac{1}{f}$$

Frequency f — SI Unit: hertz (Hz)

The number of oscillations per unit time:

$$f = \frac{1}{T}$$

$$1 \text{ Hz} = 1 \text{ cycle/s} = 1 \frac{1}{\text{s}}$$

Analogies:

- ▶ Heart beats at $f \approx 1.2 \text{ Hz} \Rightarrow T \approx 0.83 \text{ s}$
- ▶ Human hearing: 20 Hz to 20 kHz
- ▶ Medical ultrasound: $f \sim 1 \text{ MHz}$
- ▶ US power grid: $f = 60 \text{ Hz}$, $T = \frac{1}{60} \text{ s}$

Example: Ultrasound and Middle C — (cp2e_ch16_ex3)

- (a) Medical ultrasound has $T = 0.400 \mu\text{s}$. What is its frequency?
(b) Middle C on a piano has $f = 262 \text{ Hz}$. What is its period?

$$(a) \quad f = \frac{1}{T} = \frac{1}{0.400 \times 10^{-6} \text{ s}}$$

$$f = 2.50 \times 10^6 \text{ Hz} = 2.50 \text{ MHz}$$

$$(b) \quad T = \frac{1}{f} = \frac{1}{262 \text{ Hz}}$$

$$T = 3.82 \times 10^{-3} \text{ s} = 3.79 \text{ ms}$$

$f = 2.50 \text{ MHz}$ far exceeds human hearing ($< 20 \text{ kHz}$).

Ultrasound reflects off internal structures; short T means high spatial resolution.

Middle C: the speaker cone completes 264 full oscillations per second.

Check Your Understanding

A hummingbird flaps its wings 80 times per second.

(a) What is the frequency of the wing flaps?

(b) What is the period?

Come to lecture to see the answer.

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What Makes Motion “Simple Harmonic”?

Simple Harmonic Motion (SHM)

Simple harmonic motion is oscillatory motion in which the restoring force is **directly proportional to displacement**:

$$F = -k \Delta x \quad (\text{Hooke's Law})$$

Key Properties of SHM

- ▶ The period T and frequency f are **independent of amplitude X** .
- ▶ Only mass m and spring constant k determine T and f .
- ▶ Any system obeying Hooke's Law undergoes SHM: springs, pendulums (small angles), atoms in solids, ...

Period and Frequency of a Spring–Mass System

Period and Frequency of SHM (Spring–Mass)

$$T_{\text{SHO}} = 2\pi\sqrt{\frac{m}{k}} \qquad f = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$$

where m is mass (kg), k is force constant (N/m), and T is in seconds.

What Affects the Period?

- ▶ **Larger** $m \Rightarrow$ larger T (slower): more inertia to reverse.
- ▶ **Larger** $k \Rightarrow$ smaller T (faster): stronger restoring force.
- ▶ **Amplitude** X : does **not** affect T or f . (!)

$$T \propto \sqrt{m}$$

$$T \propto \frac{1}{\sqrt{k}}$$

T independent of amplitude X

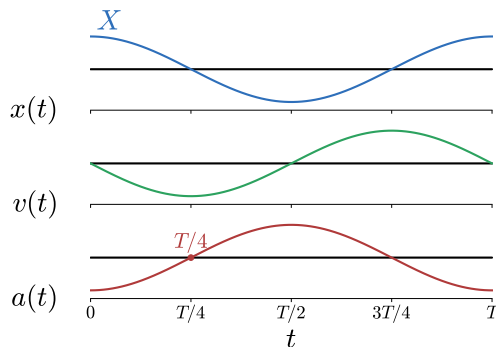
Kinematics of SHM: $x(t)$, $v(t)$, $a(t)$

If the object starts at maximum displacement X (amplitude) at $t = 0$:

$$x(t) = X \cos\left(\frac{2\pi t}{T}\right)$$

$$v(t) = -X\sqrt{\frac{k}{m}} \sin\left(\frac{2\pi t}{T}\right)$$

$$a(t) = -\frac{kX}{m} \cos\left(\frac{2\pi t}{T}\right)$$



- ▶ At $x = \pm X$ (turning points): $v = 0$, $|a|$ is maximum.
- ▶ At $x = 0$ (equilibrium): $(|v_{\max}| = X\sqrt{\frac{k}{m}})$, $a = 0$.
- ▶ x and a are always **opposite in sign**: $a = -(k/m)x$.

x , v , a are cosine/sine waves, each 90° out of phase with the next.

Example: Bad Shock Absorbers — (cp2e_ch16_ex4)

A 900 kg car has suspension springs with $k = 6.53 \times 10^4 \frac{\text{N}}{\text{m}}$.

(a) What frequency does the car bounce at? (b) What is the period?

$$\begin{aligned} \text{(a)} \quad f &= \frac{1}{2\pi} \sqrt{\frac{k}{m}} \\ &= \frac{1}{2\pi} \sqrt{\frac{6.53 \times 10^4 \frac{\text{N}}{\text{m}}}{900 \text{ kg}}} \end{aligned}$$

$$f = 1.36 \text{ Hz}$$

$$\text{(b)} \quad T = \frac{1}{f} = \frac{1}{1.36 \text{ Hz}}$$

$$T = 0.738 \text{ s}$$

$f \approx 1.36 \text{ Hz}$: the car bounces once every 0.74 s.

Same k as Example 16.1 — now we know the *bounce frequency*.

Good shock absorbers **critically damp** this motion so no bouncing occurs.

Check Your Understanding

A 0.500 kg mass is attached to a spring with $k = 8.00 \frac{\text{N}}{\text{m}}$.

(a) What is the period of oscillation?

(b) If the mass is doubled to 1.00 kg, by what factor does the period change?

Come to lecture to see the answer.

Check Your Understanding

A plucked banjo string vibrates and gradually grows quieter over several seconds.

How do the **frequency**, **period**, and **amplitude** change as the sound fades?

Come to lecture to see the answer.

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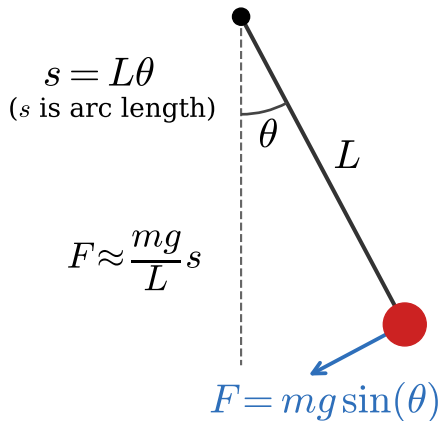
Energy in SHM

The Simple Pendulum

A **simple pendulum** is a small bob of mass m on a string of length L . For small angles ($\theta < 15^\circ$), it undergoes SHM with:

$$T_{\text{pend}} = 2\pi\sqrt{\frac{L}{g}}$$

- ▶ T depends only on L and g .
- ▶ **Mass** does not affect T . (Galileo)
- ▶ **Amplitude** does not affect T ($\theta < 15^\circ$).
- ▶ Longer pendulum $\Rightarrow$ longer period.



Pendulum $\equiv$ spring with $k = mg/L$

Example: Measuring g with a Pendulum — (cp2e_ch16_ex5)

A pendulum of length $L = 75.000$ cm has a measured period $T = 1.7357$ s.
What is the local value of g ?

Rearrange $T = 2\pi\sqrt{L/g}$:

$$T^2 = 4\pi^2 \frac{L}{g}$$

$$g = \frac{4\pi^2 L}{T^2}$$

$$= \frac{4\pi^2 (0.75000 \text{ m})}{(1.7357 \text{ s})^2}$$

$$g = 9.8281 \frac{\text{m}}{\text{s}^2}$$

This is a very precise measurement of g .

Pendulums were historically the best instrument for measuring local gravitational acceleration.

For this precision, keep $\theta_{\text{max}} < 0.5^\circ$ to stay well within the small-angle approximation.

Check Your Understanding

Two pendulums hang from the same ceiling. Pendulum 1 has a 10 kg bob; Pendulum 2 has a 100 kg bob. Both have the same length and are displaced by 12° .

- (a) Which has a longer period?
- (b) Which swings with greater maximum speed?

Come to lecture to see the answer.

Check Your Understanding

You move from sea level to a mountain city where g is slightly smaller. Your pendulum clock kept correct time at sea level.

Will it now run fast, slow, or at the same rate? What should you do to fix it?

Come to lecture to see the answer.

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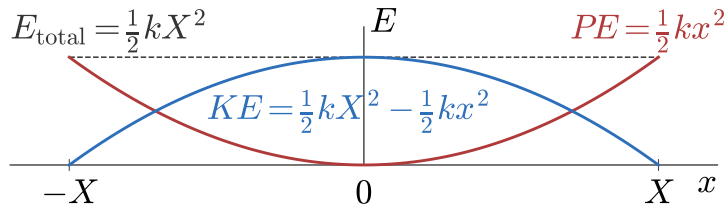
Energy in SHM

Conservation of Energy in a Simple Harmonic Oscillator (SHO)

Total mechanical energy is constant, oscillating between KE and PE:

$$KE + PE_{el} = \text{constant} \quad \Rightarrow \quad \frac{1}{2}mv^2 + \frac{1}{2}kx^2 = \frac{1}{2}kX^2$$

where X is the amplitude (maximum displacement).



- ▶ At $x = \pm X$ (turning points): $KE = 0$, $PE_{el} = \frac{1}{2}kX^2$.
- ▶ At $x = 0$ (equilibrium): $PE_{el} = 0$, $KE = \frac{1}{2}mv_{\text{max}}^2$.
- ▶ In between: energy is shared between KE and PE_{el} .

Maximum Velocity in SHM

At $x = 0$, all energy is kinetic: $\frac{1}{2}mv_{\max}^2 = \frac{1}{2}kX^2$, giving:

$$v_{\max} = X\sqrt{\frac{k}{m}}$$

At an arbitrary displacement x :

$$v = \pm\sqrt{\frac{k}{m}}\sqrt{X^2 - x^2} = \pm v_{\max}\sqrt{1 - \frac{x^2}{X^2}}$$

To increase $v_{\max}$: increase X (pull farther), increase k (stiffer spring), or decrease m .

Note: $v_{\max} = X\sqrt{k/m} = X\omega$ where $\omega = 2\pi f$ is the angular frequency.

Example: Maximum Speed of a Bouncing Car — (cp2e_ch16_ex6)

The 900 kg car with $k = 6.53 \times 10^4 \frac{\text{N}}{\text{m}}$ oscillates with amplitude $X = 0.100 \text{ m}$. What is the maximum speed of the car?

$$\begin{aligned} v_{\text{max}} &= X \sqrt{\frac{k}{m}} \\ &= (0.100 \text{ m}) \sqrt{\frac{6.53 \times 10^4 \frac{\text{N}}{\text{m}}}{900 \text{ kg}}} \end{aligned}$$

$$v_{\text{max}} = 0.852 \frac{\text{m}}{\text{s}}$$

The car reaches maximum speed passing through equilibrium (normal ride height).

$$0.852 \frac{\text{m}}{\text{s}} \approx 3.1 \frac{\text{km}}{\text{h}}$$

Not dangerous, but very uncomfortable without shock absorbers!

Check Your Understanding

A 0.200 kg mass on a spring ($k = 50.0 \frac{\text{N}}{\text{m}}$) oscillates with amplitude $X = 0.0800 \text{ m}$.

- (a) What is the maximum speed?
- (b) What is the speed when $x = 0.0400 \text{ m}$?

Come to lecture to see the answer.

Part II

Damped and Forced Oscillations

Outline of Part II: Damped and Forced Oscillations

Damped Harmonic Motion

Forced Oscillations and Resonance

Outline of Part II: Damped and Forced Oscillations

Damped Harmonic Motion

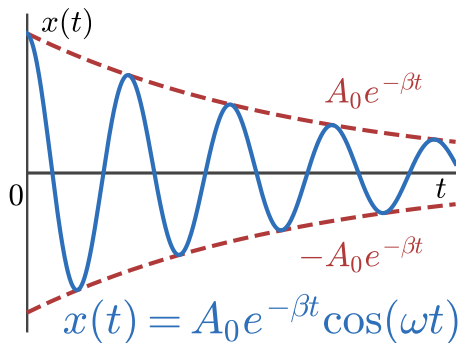
Forced Oscillations and Resonance

Damped Harmonic Motion

Damping occurs when a non-conservative force (e.g., friction, air resistance) removes mechanical energy from a harmonic oscillator:

$$W_{nc} = \Delta(KE + PE) < 0$$

- ▶ For **small** damping: period and frequency are nearly unchanged; only amplitude decreases.
- ▶ Energy is removed each cycle, so the amplitude envelope decays over time.
- ▶ **Almost all** real oscillators are damped.



Three Types of Damping

Underdamped

System oscillates and crosses equilibrium multiple times before stopping. Amplitude decays gradually. Most common in everyday oscillators.

Critically Damped

System returns to equilibrium as **quickly as possible** without oscillating. (Ideal for shock absorbers and door closers.)

Overdamped

System returns to equilibrium **without oscillating**, but more slowly than critical damping.

Example: Friction Damping a Spring — (cp2e_ch16_ex7)

A 0.200 kg mass on a spring ($k = 50.0 \frac{\text{N}}{\text{m}}$, $\mu_k = 0.0800$) is released from $X = 0.100 \text{ m}$.
(a) What is the friction force? (b) How far does it travel before stopping?

$$\begin{aligned} \text{(a)} \quad f_k &= \mu_k mg \\ &= (0.0800)(0.200 \text{ kg})(9.80 \frac{\text{m}}{\text{s}^2}) \end{aligned}$$

$$f_k = 0.157 \text{ N}$$

Using work-energy theorem with friction:

$$d = 1.59 \text{ m}$$

The spring oscillates back and forth, losing energy to friction each half-cycle.

Total path traveled = 1.59 m, though net displacement is small.

The system finally rests at $x_f = \mu_k mg/k = 3.14 \times 10^{-3} \text{ m}$ from equilibrium.

Check Your Understanding

Describe the motion of a car after hitting a pothole under each condition:

- (a) Overdamped shock absorbers
- (b) Underdamped shock absorbers
- (c) Critically damped shock absorbers

Come to lecture to see the answer.

Outline of Part II: Damped and Forced Oscillations

Damped Harmonic Motion

Forced Oscillations and Resonance

Natural Frequency and Forced Oscillations

Natural Frequency f_0

The **natural frequency** is the frequency at which a system oscillates when displaced and released with no driving force and no damping.

For a spring-mass: $f_0 = \frac{1}{2\pi} \sqrt{\frac{k}{m}}$

Forced Oscillations

When an external **periodic driving force** is applied at frequency f_{drive} , the system oscillates at f_{drive} (not f_0).

The amplitude of oscillation depends on how close f_{drive} is to f_0 .

When $f_{\text{drive}} = f_0$: **RESONANCE** — amplitude is maximized.

Resonance

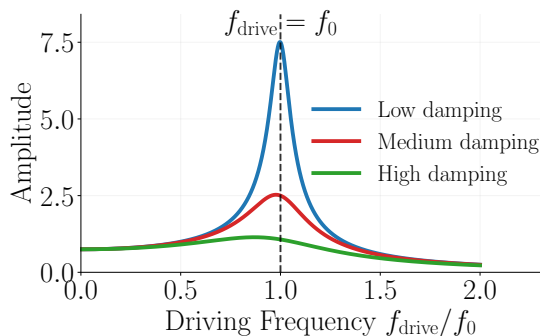
Resonance occurs when a system is driven at its natural frequency. Energy is added most efficiently, producing the **maximum amplitude response**.

Amplitude Response Curve

- ▶ Peak amplitude occurs at $f_{\text{drive}} = f_0$.
- ▶ Less damping $\Rightarrow$ sharper, taller resonance peak.
- ▶ More damping $\Rightarrow$ broader, flatter peak.

Sharp resonance: very sensitive to frequency.

Broad resonance: responds over a range of frequencies.



All curves peak at $f_{\text{drive}} = f_0$.
Less damping $\rightarrow$ taller, narrower peak.

Real-World Examples of Resonance

Resonance in Everyday Life

- ▶ **Swing:** Pushing at the natural frequency builds amplitude efficiently.
- ▶ **Radio tuning:** Adjust circuit's natural frequency to match broadcast frequency.
- ▶ **MRI:** ~ 100 MHz radio waves resonate with hydrogen nuclei in tissue.
- ▶ **Piano:** Strings resonate when a matching note is sung nearby.
- ▶ **Tacoma Narrows Bridge (1940):** Wind drove oscillations at the bridge's natural frequency $\Rightarrow$ catastrophic collapse.
- ▶ **Crystal glass:** Singing at the glass's natural frequency shatters it.

Soldiers march “route step” (out of step) across bridges to avoid exciting the bridge's natural frequency.

Check Your Understanding

A street performer sings a sustained note at a crystal wine glass. Eventually the glass shatters.

Explain using the concepts of natural frequency and resonance. What role does damping play?

Come to lecture to see the answer.

Part III

Waves

Outline of Part III: Waves

Waves

Superposition and Interference

Energy in Waves: Intensity

Outline of Part III: Waves

Waves

Superposition and Interference

Energy in Waves: Intensity

What is a Wave?

A **wave** is a disturbance that **propagates** from the place it was created, carrying energy but not matter.

Key Distinction: Energy Travels, Matter Stays

- ▶ **Energy** travels with the wave.
- ▶ **Matter** (the medium) oscillates locally but does not travel with the wave.
- ▶ Ocean waves: water molecules move in circles; the wave pattern moves across the surface.
- ▶ Sound: air molecules oscillate back and forth; the sound pulse travels forward.

Medium:

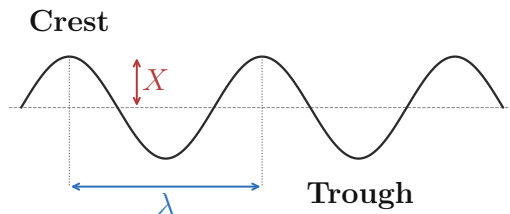
Most waves need a medium: water, air, strings, solids.

Exception: **electromagnetic waves** (light, radio, X-rays) travel through a vacuum — no medium needed.

Wave Characteristics

Wave Parameters

- ▶ **Wavelength (λ)**: distance between adjacent identical points (crest-to-crest). SI unit: meter.
- ▶ **Period (T)**: time for one complete oscillation. SI unit: second.
- ▶ **Frequency ($f = 1/T$)**: oscillations per second. SI unit: Hz.
- ▶ **Amplitude (X)**: maximum displacement from equilibrium.
- ▶ **Wave speed (v_w)**: speed at which the disturbance propagates.



Crests separated by λ ; amplitude is X ; one cycle takes time T .

v_w depends on the **medium** (material properties), not on f or λ alone.

The Wave Equation: $v_w = f\lambda$

Wave Speed — SI Unit: meter per second (m/s)

In one period T , the wave travels exactly one wavelength λ :

$$v_w = \frac{\lambda}{T} = f\lambda$$

where v_w is the wave speed, f is frequency, and λ is wavelength.

- ▶ v_w is set by the **medium**: sound in air $\approx 343 \frac{\text{m}}{\text{s}}$; sound in steel $\approx 5960 \frac{\text{m}}{\text{s}}$.
- ▶ In the same medium, increasing f **decreases** λ (wave speed is fixed).
- ▶ High-pitched sound $\Rightarrow$ shorter wavelength.

$$v_w = f\lambda$$

If v_w is fixed:

$$f \uparrow \Rightarrow \lambda \downarrow$$

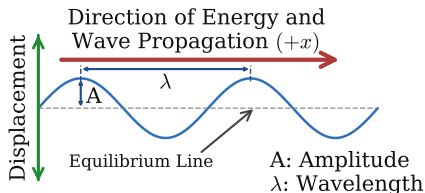
$$f \downarrow \Rightarrow \lambda \uparrow$$

Transverse vs. Longitudinal Waves

Transverse Wave

Disturbance is **perpendicular** to the direction of propagation.

Examples: waves on a string, water surface waves, light (EM waves).

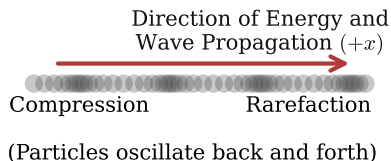


Displacement $\perp$ propagation direction.

Longitudinal Wave

Disturbance is **parallel** to the direction of propagation.

Examples: sound waves, compression waves in a spring (slinky).



Displacement $\parallel$ propagation direction.

Example: Gull in the Ocean — (cp2e_ch16_ex8)

Ocean waves with $\lambda = 10.0$ m cause a seagull to bob up and down with period $T = 5.00$ s.

What is the speed of the waves?

$$\begin{aligned}v_w &= \frac{\lambda}{T} \\&= \frac{10.0 \text{ m}}{5.00 \text{ s}}\end{aligned}$$

$$v_w = 2.00 \frac{\text{m}}{\text{s}}$$

The gull bobs up and down (oscillates) but does **not** travel horizontally with the wave.

The **wave pattern** moves at $2.00 \frac{\text{m}}{\text{s}}$; the water (and gull) stays in place on average.

Also: $f = 1/T = 0.200$ Hz

$$v_w = f\lambda = (0.200)(10.0 \text{ m}) = 2.00 \frac{\text{m}}{\text{s}} \checkmark$$

Check Your Understanding

A radio station broadcasts at $f = 101.5 \text{ MHz}$ (FM radio). Radio waves travel at the speed of light, $c = 3.00 \times 10^8 \frac{\text{m}}{\text{s}}$.

What is the wavelength of these radio waves?

Come to lecture to see the answer.

Outline of Part III: Waves

Waves

Superposition and Interference

Energy in Waves: Intensity

The Principle of Superposition

When two or more waves overlap at the same point, the total displacement is the **algebraic sum** of the individual displacements:

$$x_{\text{total}} = x_1 + x_2$$

After passing through each other, each wave continues unchanged.

Two Types of Interference

- ▶ **Constructive:** waves add together (same phase) $\Rightarrow$ larger amplitude, up to $2X$.
- ▶ **Destructive:** waves cancel (opposite phase) $\Rightarrow$ smaller or zero amplitude.

Constructive
Interference



+



Wave
amplitudes
add

Destructive
Interference



+



Wave amplitudes
cancel to give
zero amplitude

Standing Waves on a Fixed String

When two identical waves travel in opposite directions (e.g., a wave and its reflection), they interfere to form a **standing wave** with fixed **nodes** (no motion) and **antinodes** (maximum motion).

Harmonics (String Fixed at Both Ends)

Allowed frequencies for a string of length L :

$$f_n = \frac{n v_w}{2L} \quad n = 1, 2, 3, \dots$$

$n = 1$: **fundamental** (1st harmonic)

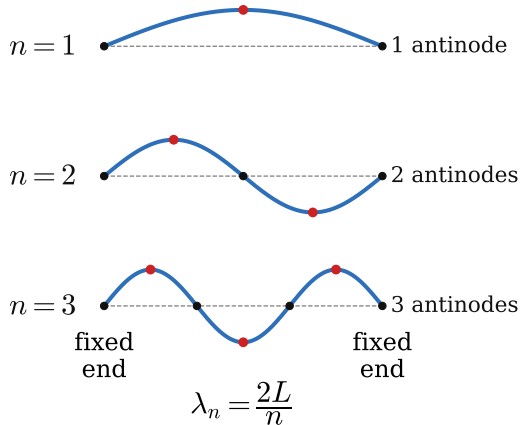
$n = 2$: 1st overtone (2nd harmonic)

$n = 3$: 2nd overtone (3rd harmonic)

(Higher harmonics: $f_n = n f_1$).



overtone $\neq$ harmonic



Nodes, Antinodes, and the Fundamental

Nodes and Antinodes

- ▶ **Node:** point of zero displacement always. String does not move.
- ▶ **Antinode:** point of maximum oscillation amplitude.
- ▶ For a fixed string with n half-waves: n antinodes, $n + 1$ nodes.

Fundamental Frequency

The lowest allowed frequency ($n = 1$):

$$f_1 = \frac{v_w}{2L}$$

n th-harmonic: $f_n = n f_1$.

Guitar String Example

$$L = 0.650 \text{ m}, v_w = 405 \frac{\text{m}}{\text{s}}$$

$$f_1 = \frac{405}{2(0.650)} = 311 \text{ Hz}$$

$$f_2 = 2f_1 = 622 \text{ Hz}$$

$$f_3 = 3f_1 = 933 \text{ Hz}$$

The fundamental sets the perceived pitch; overtones add timbre.

Beat Frequency

When two waves of **similar but not identical** frequencies superpose, they alternate between constructive and destructive interference, producing a periodic variation in amplitude called **beats**.

$$f_B = |f_1 - f_2|$$

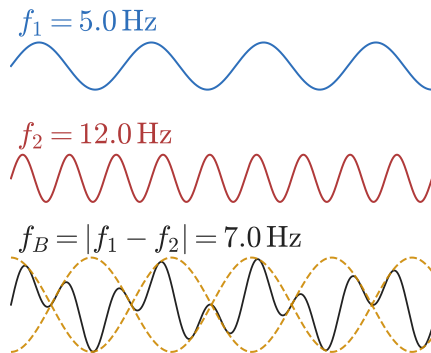
where f_B is the beat frequency (the rate at which loudness pulses).

Piano tuning: Strike a key and a tuning fork simultaneously. If beats are heard, the string is slightly off. Adjust until $f_B \rightarrow 0$ (beats disappear).

If tuning fork = 440 Hz and you hear 2 beats/s, the string is at 442 or 438 Hz.



Click here to simulate beats



Check Your Understanding

Two tuning forks are struck simultaneously: $f_1 = 440$ Hz (concert A) and $f_2 = 436$ Hz.

- (a) What beat frequency do you hear?
- (b) To tune the second fork to match the first, should you increase or decrease f_2 ? How will you know when it is in tune?

Come to lecture to see the answer.

Check Your Understanding

You set up a stereo system and notice one corner of the room sounds dull while another area is excessively loud.

Explain these effects using the principle of superposition.

Come to lecture to see the answer.

Outline of Part III: Waves

Waves

Superposition and Interference

Energy in Waves: Intensity

Wave Intensity

Intensity I — SI Unit: watt per square meter (W/m^2)

Intensity is the power per unit area carried by a wave:

$$I = \frac{P}{A}$$

where P is power (W) and A is the area (m^2) through which the wave passes.

- ▶ $I \propto X^2$ (amplitude squared): doubling amplitude $\Rightarrow$ **four times** the intensity.
- ▶ Energy delivered: $E = I \cdot A \cdot t$.
- ▶ Point source radiating equally:
 $I = P/(4\pi r^2)$ — intensity falls as $1/r^2$.

$$I = \frac{P}{A}$$

$$E = I \cdot A \cdot t$$

Same total power, smaller area $\rightarrow$
higher intensity.

Example: Solar Energy Collector — (cp2e_ch16_ex9)

Solar intensity: $I = 700 \frac{\text{W}}{\text{m}^2}$. A solar panel has $A = 0.500 \text{ m}^2$.

(a) How much energy is collected in $t = 4.00 \text{ h}$?

(b) A lens concentrates sunlight onto an area 200 times smaller. What is the new intensity?

$$(a) \quad E = I \cdot A \cdot t$$

$$= (700 \frac{\text{W}}{\text{m}^2})(0.500 \text{ m}^2)(4.00 \times 3600 \text{ s})$$

$$E = 5.04 \times 10^6 \text{ J}$$

$$(b) \quad I' = \frac{P}{A/200} = 200 \frac{P}{A} = 200 I$$

$$I' = 1.40 \times 10^5 \frac{\text{W}}{\text{m}^2}$$

Part (a): $\approx 5 \text{ MJ}$ — enough to run a microwave for over an hour.

Part (b): $1.40 \times 10^5 \frac{\text{W}}{\text{m}^2}$ is intense enough to ignite paper instantly.

Same total power P — just concentrated onto a much smaller area.

Intensity and Constructive Interference

Why Constructive Interference Quadruples Intensity

When two identical waves of amplitude X and intensity I_0 interfere constructively:

$$X_{\text{total}} = 2X \quad \Rightarrow \quad I_{\text{total}} = 4 I_0 \quad (\text{not } 2I_0!)$$

because $I \propto X^2$: doubling amplitude **quadruples** intensity.

Two $1.00 \frac{\text{W}}{\text{m}^2}$ waves interfere constructively:

$$I_{\text{total}} = 4 \times 1.00 \frac{\text{W}}{\text{m}^2} = \boxed{4.00 \frac{\text{W}}{\text{m}^2}}$$

Energy is *redistributed*, not created: bright spots are extra bright, dark spots have zero intensity.

Constructive: $I = 4 I_0$

Destructive: $I = 0$

Average: $I = 2 I_0$ (energy conserved)

Check Your Understanding

A speaker produces intensity I_0 at a distance of 2.0 m.

(a) Which property of the wave most directly determines its intensity?

(b) You move to 4.0 m away (double the distance). What happens to I ? (Use $I \propto 1/r^2$.)

Come to lecture to see the answer.

Chapter Summary

Equation	Variables	Conditions
$F = -k\Delta x$	k : force constant; Δx : displacement	Hooke's Law
$PE_{el} = \frac{1}{2}kx^2$	elastic potential energy stored	spring displaced by x
$f = 1/T$	f : frequency; T : period	always
$T_{SHO} = 2\pi\sqrt{m/k}$	m : mass; k : spring constant	spring-mass SHM
$v_{\max} = X\sqrt{k/m}$	X : amplitude	max speed in SHM
$\frac{1}{2}mv^2 + \frac{1}{2}kx^2 = \frac{1}{2}kX^2$	energy conservation in SHM	undamped SHM
$T_{\text{pend}} = 2\pi\sqrt{L/g}$	L : length; g : gravitational accel.	small-angle pendulum
$v_w = f\lambda$	v_w : wave speed; λ : wavelength	all waves
$f_B = f_1 - f_2 $	f_B : beat frequency	superposition of close f
$f_n = nv_w/(2L)$	$n = 1, 2, 3, \dots$; L : string length	standing waves, fixed ends
$I = P/A$	P : power; A : area	wave intensity